Python Program

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# Aim: Implement a Hash Table using Division Method
# and Linear Probing

# Create a fixed-size hash table
m = int(input("Enter the size of hash table: "))
hash_table = [None] * m
DELETED = "DELETED"

# 1. Insert Key
def insert(key):
    index = key % m
    start = index

    while hash_table[index] is not None and hash_table[index] != DELETED:
        if hash_table[index] == key:
            print("Key already exists.")
            return
        index = (index + 1) % m

        if index == start:
            print("Hash table is full.")
            return

    hash_table[index] = key
    print("Key inserted successfully.")
```

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# 2. Search Key
def search(key):
    index = key % m
    start = index

    while hash_table[index] is not None:
        if hash_table[index] == key:
            print("Key Found at index", index)
            return

        index = (index + 1) % m

        if index == start:
            break

    print("Key not found")

# 3. Delete Key
def delete(key):
    index = key % m
    start = index

    while hash_table[index] is not None:
        if hash_table[index] == key:
            hash_table[index] = DELETED
            print("Key deleted successfully.")
            return
```

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        index = (index + 1) % m

        if index == start:
            break

    print("Key not found")

# 4. Display Hash Table
def display():
    print("\nHash Table:")
    for i in range(m):
        if hash_table[i] is None:
            print(i, ":", "Empty")
        elif hash_table[i] == DELETED:
            print(i, ":", "Deleted")
        else:
            print(i, ":", hash_table[i])


# Main Menu
while True:
    print("\n--- Hash Table Menu ---")
    print("1. Insert Key")
    print("2. Search Key")
    print("3. Delete Key")
    print("4. Display Hash Table")
    print("5. Exit")
```

```
choice = input("Enter your choice: ")

if choice == "1":
    key = int(input("Enter key to insert: "))
    insert(key)

elif choice == "2":
    key = int(input("Enter key to search: "))
    search(key)

elif choice == "3":
    key = int(input("Enter key to delete: "))
    delete(key)

elif choice == "4":
    display()

elif choice == "5":
    print("Exiting program.")
    break

else:
    print("Invalid choice.")
```

Sample Output

Enter the size of hash table: 5

--- Hash Table Menu ---
1. Insert Key
2. Search Key
3. Delete Key
4. Display Hash Table
5. Exit
Enter your choice: 1
Enter key to insert: 10
Key inserted successfully.

--- Hash Table Menu ---
1. Insert Key
2. Search Key
3. Delete Key
4. Display Hash Table
5. Exit
Enter your choice: 1
Enter key to insert: 15
Key inserted successfully.

--- Hash Table Menu ---
1. Insert Key
2. Search Key
3. Delete Key
4. Display Hash Table
5. Exit
Enter your choice: 1
Enter key to insert: 20
Key inserted successfully.

--- Hash Table Menu ---
1. Insert Key
2. Search Key
3. Delete Key
4. Display Hash Table
5. Exit
Enter your choice: 4

Hash Table:
0 : 10
1 : 15
2 : 20
3 : Empty

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4 : Empty

--- Hash Table Menu ---
1. Insert Key
2. Search Key
3. Delete Key
4. Display Hash Table
5. Exit
Enter your choice: 2
Enter key to search: 15
Key Found at index 1

--- Hash Table Menu ---
1. Insert Key
2. Search Key
3. Delete Key
4. Display Hash Table
5. Exit
Enter your choice: 3
Enter key to delete: 15
Key deleted successfully.

--- Hash Table Menu ---
1. Insert Key
2. Search Key
3. Delete Key
4. Display Hash Table
5. Exit
Enter your choice: 4

Hash Table:
0 : 10
1 : Deleted
2 : 20
3 : Empty
4 : Empty

--- Hash Table Menu ---
1. Insert Key
2. Search Key
3. Delete Key
4. Display Hash Table
5. Exit
Enter your choice: 5
Exiting program.
```